pandas – SQL



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“psycopg2” vs "sql-alchemy"

- **psycopg2**
  - **Low-level** PostgreSQL driver (DB-API 2.0).
  - You write **raw SQL** and manage connections, cursors, commits, rollbacks yourself.
- Example:



Example: using “psycopg2”

```
# imports
import psycopg2

# establish db connection
conn = psycopg2.connect(database="test", user="test", password="test",\
                        host="10.100.71.21", port="5432")
print("Database connected successfully")

cur = conn.cursor()
# set schema to "company3"
cur.execute("SET search_path TO company3")

sql = "SELECT e.ename as emp_name, s.ename as supervisor_name "\
    + "FROM EMPLOYEE AS e JOIN Employee AS s ON (e.super_eno=s.eno) WHERE e.dno=4"

cur.execute( sql ) # returns "None"
rows = cur.fetchall() # returns "list" object
for row in rows: # row is tuple object
    print( row[0], row[1] )

cur.close()
conn.close()
```

psycopg2

- Pros:
  - Lightweight,
  - fast,
  - fewer dependencies.
  - Full control over PostgreSQL-specific features.
  - Useful if you only target PostgreSQL and don’t need abstraction.
- Cons:
  - You manage boilerplate (connections, transactions).
  - Harder to switch databases.
  - No built-in ORM.

SQL-Alchemy

- **Abstraction layer** on top of drivers like psycopg2 (it actually uses psycopg2 under the hood for PostgreSQL).
- Provides:
  - **Connection pooling** (efficient reuse of DB connections).
  - **Unified interface** across many databases (Postgres, MySQL, SQLite, Oracle, etc.).
  - **Core**: SQL expression language (build queries in Python objects, not just strings).
  - **ORM**: Object-Relational Mapping (map Python classes ↔ DB tables).

SQLAlchemy

- Pros:
  - Cleaner integration with Pandas (read_sql, to_sql).
  - Built-in connection pooling.
  - Database-agnostic (switch DBs by changing the URL).
  - Optional ORM layer for complex apps.
  - Safer query construction with sqlalchemy.text() (prevents SQL injection).
- Cons:
  - Slightly more overhead than using psycopg2 directly.
  - For very performance-sensitive apps, psycopg2-only can be leaner



Example: using “SQL-Alchemy”